THE UNITED REPUBLIC OF TANZANIA

PRESIDENTS' OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

SINGIDA REGION

FORM FOUR MOCK EXAMINATION

032/2A CHEMISTRY 2A (ACTUAL PRACTICAL)

Time 2:30 Hours 28th July, 2021 a.m

Instructions

1. This paper consists of two (2) questions carrying 25marks each. Answer all questions.
2. A non programmable calculator and qualitative guidance pamphlet can be used after thoroughly checked by supervisor.
3. Write your examination number on every page of your answer booklets.
4. Cellular phones and any other unauthorized materials are NOT allowed in the examination room.
5. The following constants may be used:
   Atomic masses: H=1, C=12, K=39, Cl=35.5,Na=23
   GMV at STP= 22.4dm^3
   1litre= 1dm^3=1000cm^3

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1. You are provided with the following solutions:
   AA: Containing 0.1 mole Hydrochloric acid per dm^3 of solution.
   BB: Containing 2.65g X_2CO_3 per 0.5dm3 of solution.
   Methyl orange indicator (MO).

   Questions
   (a) Titrate AA (in burette) against BB (in a conical flask) using two drops of your indicator and obtain three titre values. Record your data in tabular form.
   (b) (i) .....cm3 of BB required ..... cm3 of AA for complete reaction.
   (ii) Write a balanced chemical equation between BB and AA and the corresponding ionic equation with the state symbols.
   (iii) Showing your procedures clearly, calculate the molar mass of X_2CO_3 and show how to identify X.

2. You are provided with the following:
   T1: A solution containing 79g of sodium thiosulphate in one litre.
   T2: A solution containing 0.1mol/dm3 Hydrochloric acid.
   T3: Distilled water.
   Plain paper marked X.

   Procedures

   (i) Place a 100cm^3 beaker on top of letter X on a plain paper provided
   (ii) Measure 10cm^3 of T_1 and put it in a beaker in (i)
   (iii) Measure 10cm^3 of T_2 and put into a beaker containing T_1 and immediately start the stop watch and observe the changes from above.
   (iv) Record the time taken for the disappearance of letter X.
   (v) Repeat experiment 2 to 5 using the data shown in table 1.

   Table1

Number of experiments	Volume of $T_2(cm^3)$	Volume of $T_2(cm^3)$	Volume of $T_3(cm^3)$	Time(t) in seconds	$1/t(sec^{-1})$
1	10	10	0		
2	10	8	2		
3	10	6	4		
4	10	4	6		
5	10	2	8		

   Questions
   (a) What is the aim of the whole experiment?

(b) Complete table 1
(c) Give the reasons, identify the experiment in which the reaction was
    (i) Fast
    (ii) Slow
(d) With state symbols, write the balanced chemical equation for the reaction between T1 and T2
(e) List four factors which affect the rate of chemical reaction.
(f) Write the electronic configuration of the product which causes the solution to cloud letter X.
(g) What can you conclude from the data obtained?

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CHEMISTRY 2A(ACTUAL PRACTICAL)

MARKING SCHEME

1. (a) Volumetric analysis

Table of results 06Marks

Volume of acid is 20cm3 for pipette of 20 and 25cm3 for pipette of 25cm3

(b) (i) $20cm^3$ or $25cm^3$ of BB required $20cm^3$ or $25cm^3$ +/-5 of (02marks)

(ii) $X_2CO_{3(aq)} + 2HCl_{(aq)} \longrightarrow 2XCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$ (3marks)

$2X^+_{(aq)} + CO^{2+}_{3(aq)} + 2H^+_{(aq)} + 2Cl^-_{(aq)} \longrightarrow 2X^+_{(aq)} + 2Cl^-_{(aq)} + H_2O_{(l)} + CO_{2(g)}$

Net ionic equation

$CO_{3(aq)}^{2-} + 2H^+_{(aq)} + \longrightarrow H_2O_{(l)} + CO_{3(aq)}$

(iii)Data given

Molarity of acid=0.1M

Molarity of base=?

Na=2

Nb=1

Volume of acid=$20cm^3$

Volume of base=$20cm^3$

Mass of x_2co_3=2.65g in $0.5dm^3$

$\frac{MaVa}{MbVb} = \frac{\eta a}{\eta b}$ (1 mark)

Mb = $\frac{MaVa\ x\ \eta b}{Vb\ x\ \eta a} = \frac{0.1\ x\ 20\ x\ 1}{20\ x\ 2} = 0.05 Mol/dm^{-3}$ (0.5 mark)

From

Molarity = $\frac{concentration}{molar\ mass}$ (0.5 mark)

From

Concentration = $\frac{mass}{volume}$ (0.5 mark)

Conc = $\frac{2.65g}{0.5dm^3} = 5.3gdm^3$ (0.5 mark)

From,

Molar mass = $\frac{conc}{molarity}$ (0.5 mark)

$= \frac{5.3gdm^{-3}}{0.05\ mol/dm^{-3}} = 106g/mol$

Molar mass of X_2CO_3 is106g/mol (1 mark)

X_2CO_3 = 106 (1 mark)

2X + 12 + (16 x3) = 106 (1 mark)

2X = 106 – 60 (1 mark)

. $\frac{2x}{2} = \frac{46}{2}$ (1 mark)

X = 23 (1 mark)

2.

a) To identify the effects of concentration in the rate of chemical reaction

b)

Time (t) in seconds	$\frac{1}{t}\ (sec^{-1})$
27	0.037
30	0.033
36	0.028
60	0.0167
120	0.0083

c)

i. It was fast due to high concentration of T_1 and T_2 (3marks)

ii. It was slow because the conc of T_1 and T_2 was lowered by addition of H_2O (3marks)

d) $HCL_{(aq)} + Na_2S_2O_{3(aq)} \longrightarrow NaCL_{(aq)} + H_2O_{(l)} + SO_{2\,(g)} + S_{(s)}$ (3 marks)

e)

i. Concentration

ii. Temperature

iii. Pressure

iv. Surface area

v. Catalyst

f) $_{16}S$ = 2:8:6 (3 marks)

g) Decrease in concentration will decrease rate chemical reaction and increase in concentration will increase in chemical reaction (4 marks)